THE LAWS OF SOFTWARE TEAMS

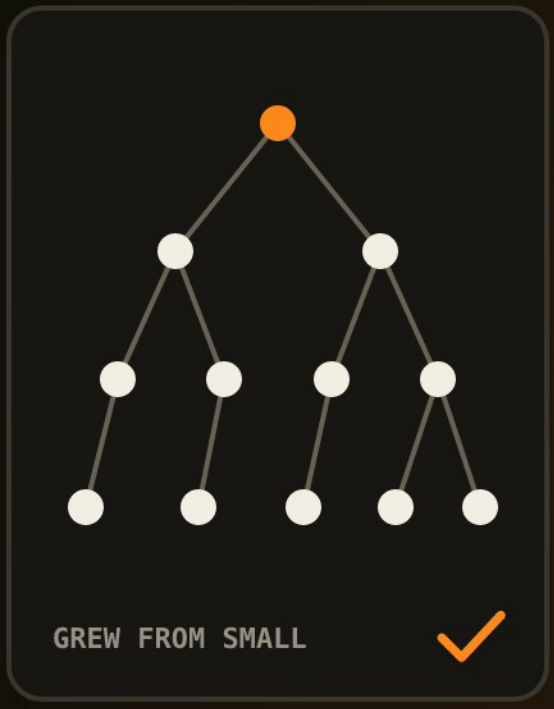
Gall's law

A complex system that works grew from a simple one that worked

swipe →

Ivan Rykov

What works started small



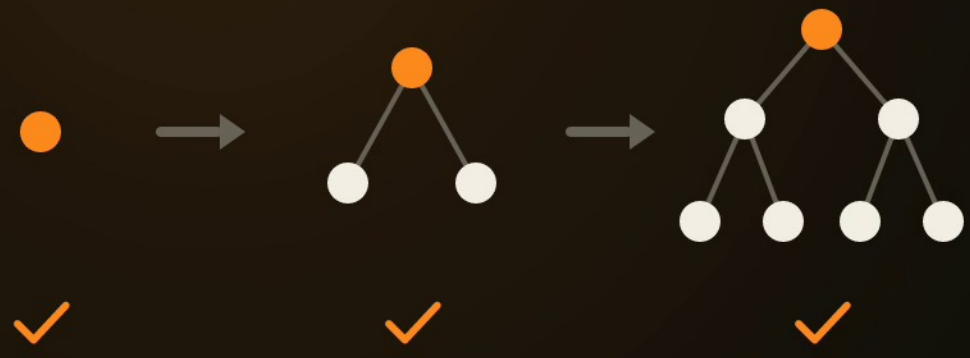
Written down in 1975 by a children's doctor

illustration

"A complex system that works is invariably found to have evolved from a simple system that worked"

John Gall · General Systemantics · 1975

a paediatrician who wrote about systems



each stage works before the next one · illustration

Designed whole, never works

> "A complex system designed from scratch never works and cannot be patched up to make it work. You have to start over with a working simple system"
>
> John Gall, Systemantics, 1975, p. 71



the big design + patches



start over, small

Linux: just a hobby

COMP.OS.MINIX · 25 AUGUST 1991

From: Linus Torvalds

“just a hobby, won’t be big and professional like gnu”



10,000

1991 · version 0.01

23.3M+

lines

2018 · version 4.15

circle area shows lines of code · 2018 count without comments

05 REWRITTEN

Netscape: the full rewrite

rewrite from scratch

4.0	~~5.0~~	6.0
1997	never shipped	beta · April 2000

almost three years

JOEL SPOLSKY · APRIL 2000

“the single worst strategic mistake that any software company can make”

Things You Should Never Do, Part I

Nobody knows every requirement

KNOWN ON DAY ONE

- sign in
- create an invoice
- send it

A working system shows you what it really needs

FOUND BY USING IT

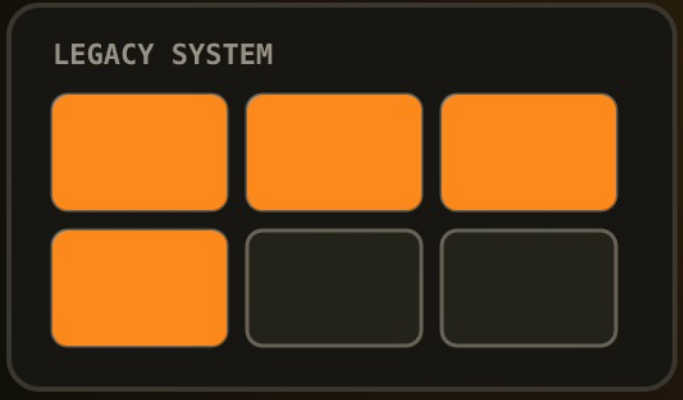
- time zones
- retries
- bulk edit
- audit log
- rate limits
- partial refunds
- export to CSV

illustration

Start small
Replace in pieces

UI | API | DB | deploy

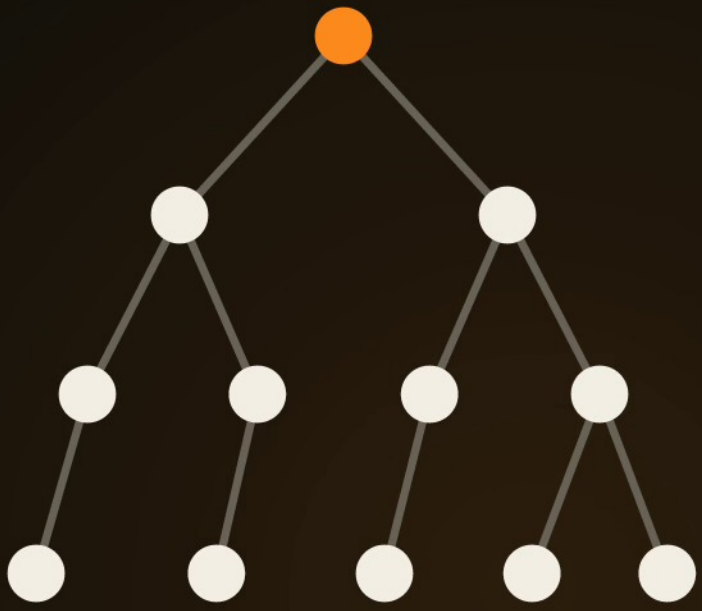
the smallest version that works end to end — then grow it



new pieces
around the old,
one at a time

the strangler fig pattern · Martin Fowler

Grow systems
Do not design them whole



TAKEAWAY

Grow systems
Do not design them whole

Sources: John Gall, Systemantics, 1975 · Joel Spolsky, 2000 · Martin Fowler

next: Hyrum's law